Modelling Ocean Fluid Motion in Ocean Structures

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Abstract

Context: The University of Canterbury Computer Vision Department has created an underwater robot capable of collecting mussels from dropper lines in mussel farms. However, due to the complex nature of the ocean, stabilising the robot against the movement of the dropper lines is difficult. This makes sample collection difficult as the robot moves around in the currents. **Objectives:** The purpose of this research project is to investigate methods of modelling these ocean fluids in structured environments, such as mussel farms. Using this information, a model will be developed that can be used to aid in stabilising the robot during operation. **Method:** This will be completed in three work streams: an initial research phase will investigate existing models and the effects that underwater structures have on fluid flows. From there, a model will be developed, simulated, and refined until a suitable level of accuracy compared to the real world is achieved. Finally, the model will be used to develop a control system for use on the robot to help improve its operational stability and thus make collecting mussels easier.

1 Introduction

1.1 Context

The University of Canterbury (UC) Computer Vision Department has been working closely with mussel farms to develop an automated remotely operated vehicle (ROV), see Figure 1, capable of retrieving a sample of mussels from mussel farm drop lines before harvest. This is currently performed using a crane mounted on a large boat to lift the drop lines out of the water. This is costly to run and takes considerable man-hours to retrieve the necessary samples. The ROV would allow a much smaller boat and fewer staff to be used to collect the samples, drastically reducing costs for the mussel farmers. Thus, this has been a high-interest project for these mussel farmers who would benefit from the reduction in expenses throughout their operation.

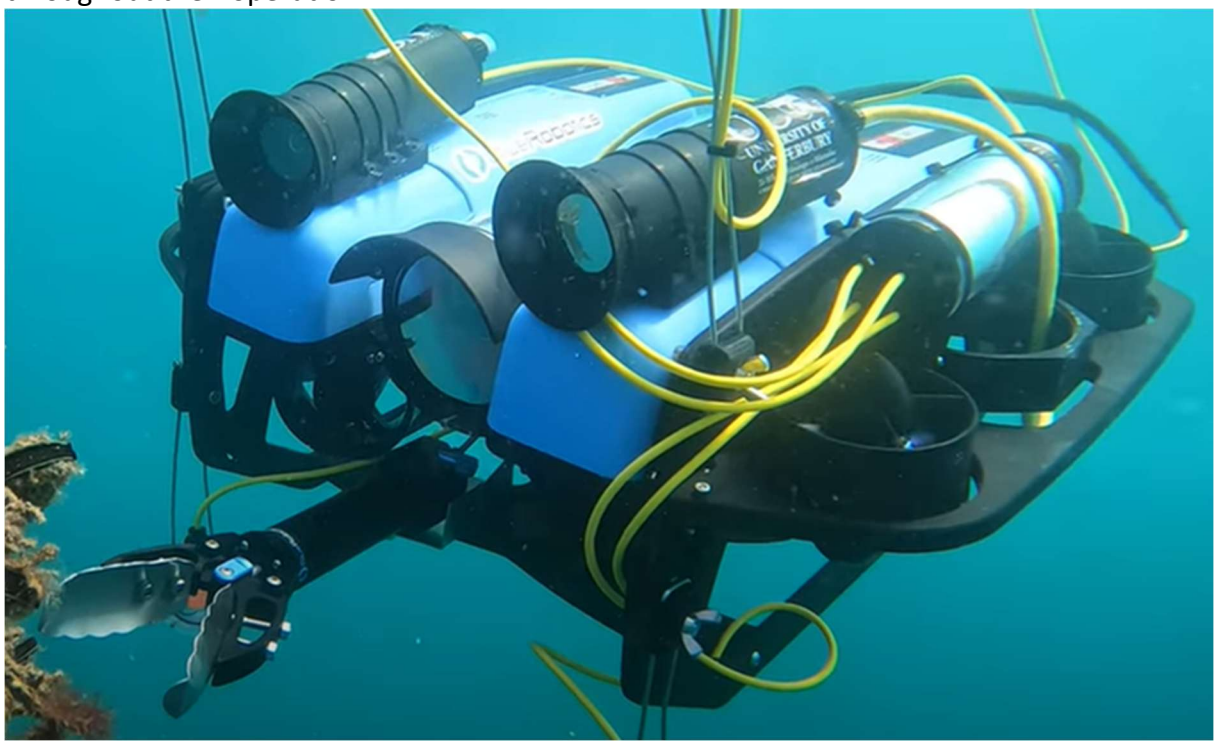


Figure 1: UC Vision Mussel Rope Scanning ROV. rom: https://ucvision.org.nz/aquaculture/

1.2 Area Under Examination

Presently, the ROV is functional. It can navigate in the ocean in all directions, identifying mussels on the drop lines, and can grab mussels under manual operation. However, for the ROV to be a feasible replacement for the current crane system, it needs to be able to automatically align and grab the mussels off the line, as manual alignment is highly tedious and can take considerable time to retrieve just one mussel.

This research project aims to investigate modelling the fluid motion in ocean/water-based structures, such as mussel farms. Specific investigations will be conducted to determine contributing factors and work towards developing the final model. These investigations will look into pre-existing models, how the structures impact fluid dynamics, and how measurable factors can be used to develop a controllable model.

2 Literature Review

2.1 Mussel Farms and Infrastructure

Mussel farming is a major component of international seafood operations and is an environmentally stable form of seafood farming. The New Zealand mussel market largely focuses on the production of the green-lipped mussel. Production of these is largely based in regions such as the Marlborough Sounds, Tasman Bay, and the Coromandel Peninsula, as these locations have the right conditions to sustain consistent shellfish growth. Mussel farming is most practised using a suspended longline system where multiple pre-seeded dropper lines are affixed to a single backbone rope, which is suspended between large buoys at each end and then anchored to the seabed. The vertical dropper lines provide the base for the mussels to attach and grow, allowing for easy management and inspection throughout their lifecycle [3].

As mussels begin to grow from the seeded droppers, the increased mass introduces significant drag forces which influence the movement of water within the farm. Research shows that suspended shellfish structures can reduce water currents and generate wakes that can change circulation patterns. It has been shown that mussel droppers change the flow behaviour by introducing resistance to currents both inside and around these shellfish farms [10]. These resistive effects influence the nutrient transport, biological productivity, and the sedimentation process of the seabed below. The impacts also extend beyond simply the environment, as current and flow changes can affect the stability of underwater vehicles. As seafood farming operations grow, automated monitoring technologies are being considered more heavily. Robotic systems capable of operating within this complex environment could provide improved monitoring of mussel growth and farm infrastructure while reducing the reliance and expense of current manual inspection methods.

2.2 Use of ROVs for Mussel Farm Inspection

ROVs have been widely established as a tool for inspections and monitoring in underwater environments. The ROV platform allows for a wide variety of data collection without the need for human divers or large vessel infrastructure. This reduces risks and costs for operators in these areas while also allowing for more frequent data collection and a higher level of consistency of the collected data. ROVs are particularly well-suited to inspection tasks because they can hover in place, carrying sensors such as cameras, sonar systems, and manipulators [2]. At present, these underwater ROVs are mostly used by offshore industries to inspect underwater infrastructure such as pipelines, cables, and structural components. Similar technologies are being explored by the seafood sector. In mussel farming operations, inspections are traditionally performed by either divers or boat-mounted cranes, which allows for a visual assessment of droplines and the collection of samples to monitor growth and health. This is expensive, labour-intensive and limited by weather conditions, water depth, and diver safety constraints.

Utilising ROVs for inspections and sampling offers the potential to automate these tasks. Underwater ROVs equipped with cameras and computer vision systems can assess mussel density, growth rates, and environmental conditions while simultaneously reducing operational costs and human risk. However, these robots must operate near the dropper lines

whilst capturing clear imagery for analysis. This creates its own challenges, as environmental disturbances such as water currents, turbidity, and aquatic life can all impact image quality. Effective stabilisation capabilities are a key requirement for underwater ROVs.

2.3 Stabilisation and Control of Underwater Robots

The complex dynamic fluid flows underwater make the autonomous motion of an ROV highly challenging. Forces such as drag, buoyancy, and applied current forces all impact the motion of the robot in all six axes. These impacts are typically non-linear and thus difficult to predict and combat without complex analysis of the environment. Thus, accurate motion control requires robust control systems capable of compensating for these forces while maintaining desired positions or trajectories [4].

Many ROVs rely on closed-loop control systems such as proportional integral derivative (PID) controllers because of their simplicity and reliability. However, more advanced control systems have been developed to improve control performance in dynamic environments. These include sliding mode control, adaptive control, and model predictive control; all have been investigated to enhance the rejection of disturbances and track the trajectory of the ROV [1]. These systems use feedback from onboard sensors, including inertial measurement units (IMUs), depth sensors, and acoustic or light positioning systems such as sonar or lidar. Additionally, modelling these complex systems is an important part of underwater ROV control design. Mathematical models describing the interaction between the vehicle and the surrounding fluid allow engineers to predict how the ROV will respond to environmental influences such as currents. These models allow simulations to be performed to rapidly prototype and refine control systems for these underwater ROVs so they can move effectively underwater despite changing system conditions.

2.4 Modelling Fluid Motion in Ocean Structures

Modelling fluid motion in structured environments such as mussel farms is essential for understanding the forces acting on underwater vehicles. Unlike open-water conditions, mussel farms are a dense, complicated environment of dropper lines that interact with background currents, creating complex flows. These interactions generate turbulence, wakes, and variations in flow velocity, resulting in a dynamic, non-uniform system [10]. Such conditions can introduce unpredictable disturbances that affect the stability and control of ROVs.

In other studies, a range of modelling approaches have been developed to represent these flow behaviours. Simplified drag-based models are commonly used to approximate the resistive effects of droplines using drag coefficients [6]. These models are efficient and suitable for large-scale simulations but lack accuracy with local turbulence. More advanced methods, such as computational fluid dynamics (CFD), can provide detailed representations of flow interactions and wake structures within mussel farms. However, these approaches are often computationally expensive and less practical for real-time applications. For ROV stabilisation, the model must balance accuracy and computational efficiency. Reduced-order models capturing the dominant flows and frequencies can be used to predict environmental forces acting on the vehicle. This will allow for an improved control system to be developed. Thus, improving the performance of the ROV at stabilising against dropper lines.

3 Methods and Materials

3.1 Research Objectives

The purpose of this proposal is to investigate and model fluid motion in ocean structures such as mussel farms. This is being done as a tangential research project to an underwater robot project, for which the findings of this research will be implemented as a stabilisation solution. This research project will aim to determine the feasibility of modelling these complex fluid systems, and if determined to be feasible, develop a custom model for use in the underwater ROV. The model will need to be provably accurate and computationally lightweight. If either of these objectives is not met, the model would not be suitable for implementation in the ROV.

3.2 Approach

In order to achieve the aforementioned goals, multiple work streams will be undertaken. These will be as follows:

3.2.1 Research into Preexisting Solutions

An initial investigation into the preexisting solutions will be performed. This will explore topics such as fluid modelling and how environmental factors (such as dropper lines) impact the fluid flow. This can then be compared against existing ocean models to determine approaches and techniques that can be utilised to develop the model.

3.2.2 Initial Implementations and Simulation

After initial research has been conducted. First iterations of the model can be developed using techniques and knowledge gained from the research phase. The model should be realisable using simulation software in some manner such that its accuracy can be assessed to determine if it is fit for use on the ROV. Additionally, iterations and improvements will be made to the model throughout this phase.

3.2.3 Refinements and Real-World Testing

After an initial model has been developed that satisfies the criteria for implementation. This model can be translated into a control system for implementation on the ROV. This will allow for real-world testing and additional refinement. While the implementation of the model is not necessarily a requirement for this research project, it will enable additional testing and refinement of the model to ensure the best accuracy is achieved.

Declarations

- *AI usage*: ChatGPT was utilised to draft the literature review talking points.

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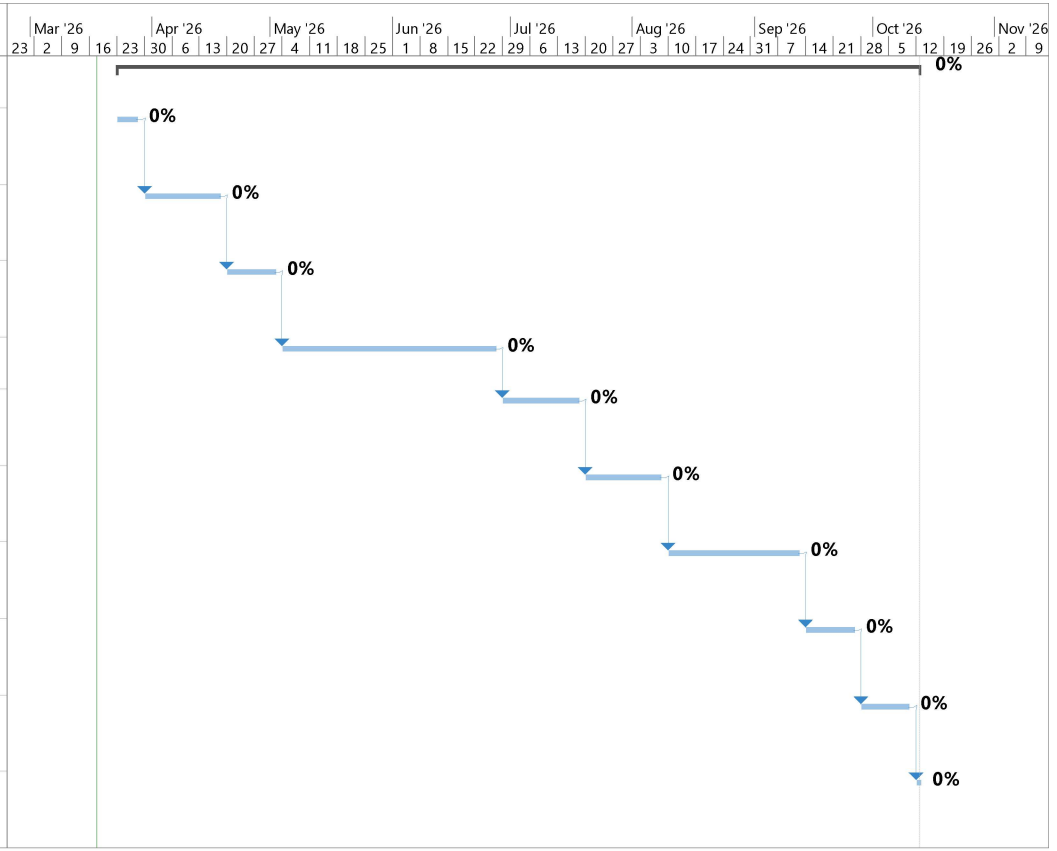
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Appendix A: Project planning

ID	WBS	Task Mode	Task Name	Duration	Start	Finish	Cost	Resource Names
1	**1**		**Ocen Fluid Modelling**	**146 days**	**Mon 23/03/26**	**Mon 12/10/26**	**$0.00**	**Alex**
2	1.1		Communicate with project technicians to gather current data and performance	5 days	Mon 23/03/26	Fri 27/03/26	$0.00	Alex
3	1.2		Investigate possible solutions	15 days	Mon 30/03/26	Fri 17/04/26	$0.00	Alex
4	1.3		Establish performance of existing systems in simulation software	10 days	Mon 20/04/26	Fri 1/05/26	$0.00	Alex
5	1.4		Develop initial model	40 days	Mon 4/05/26	Fri 26/06/26	$0.00	Alex
6	1.5		Test initial model with simulaton	15 days	Mon 29/06/26	Fri 17/07/26	$0.00	Alex
7	1.6		Iterate upon model until suitable accuracy achieved	15 days	Mon 20/07/26	Fri 7/08/26	$0.00	Alex
8	1.7		Implement model as control system on ROV	25 days	Mon 10/08/26	Fri 11/09/26	$0.00	Alex
9	1.8		Real world testing and further iteration	10 days	Mon 14/09/26	Fri 25/09/26	$0.00	Alex
10	1.9		Formalise findings	10 days	Mon 28/09/26	Fri 9/10/26	$0.00	Alex
11	1.10		Present solution and findings.	1 day	Mon 12/10/26	Mon 12/10/26	$0.00	Alex



Appendix B: H&S hazard assessment

Note: The Risk matrix identifies risks involved with the operation of the ROV in later testing.

Risks of system in its CURRENT STATE, with its existing controls.								Risks of system in its FUTURE STATE after these treatments. Insert any new threats caused by the treatments.			
Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Sensors	Reads incorrect data	2	3	6	Minimal action required, code alterations	Alex	No	2	3	6	Yes - Further action not neccessary
Cameras	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Cameras	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Battery	Overheating	3	2	6	Check battery health/condition prior to operation and monitor power draw while in use	Alex	No	3	2	6	Yes - Further action not neccessary
Battery	Electric Fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Battery	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Propellers	Collision with object/ wildlife	3	4	12	Monitor surroundings, Ensure personal safety, before startup inspect propellers	Alex	Yes	3	3	9	Yes - Further action not neccessary
Propellers	Mechanical failure	3	2	6	Inspect propellors prior to use of ROV	Alex	No	3	2	6	Yes - Further action not neccessary
Motors	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Motors	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Motors	Overheating	3	2	6	Monitor power draw from motors while ROV in use	Alex	Yes	3	2	6	Yes - Further action not neccessary
Motors	Noise – environmental damage	2	4	8	Ensure motors don't result in propellor cavtation - monitor rpms	Alex	Yes	2	4	8	Yes - Further action not neccessary
LEDs	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
LEDs	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
LEDs	Bright lights	3	4	12	Proper PPE, Disable LED when it is out of the water	Alex	Yes	2	3	6	Yes - Further action not neccessary
Controllable Underwater Events	Collision – self inflicted	2	4	8	Montitor surrounding while operating and approach motion controll sensibly	Alex	Yes	2	4	8	Yes - Further action not neccessary
Uncontrollable Underwater Events	Strong current	2	3	6	Test ocean currents prior to operation and set current threshhold for non-operation	Alex	Yes	2	3	6	Yes - Further action not neccessary
Uncontrollable Underwater Events	Encounter with wildlife	4	4	16	Monitor surroundings and ensure no abundance of wildlife (fish, dolphins etc) in testing area	Alex	Yes	4	4	16	NO – not much we can do to mitigate this
Uncontrollable Underwater Events	Collision	2	3	6	Monitor surroundings and motion of robot while in operation and actively avoid collisions through piloting robot	Alex	Yes	2	3	6	Yes - Further action not neccessary